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20 May 1971
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14 June 1963

MILITARY SPECIFICATION

DETECTOR, MAGNETIC AZIMUTH DSU-4/A, HIGH TEMPERATURE

1. SCOPE

1.1 This specification covers the requirements for a magnetic azimuth detector, designated DSU-4/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Specifications

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PP-B-601	Boxes, Wood, Cleated-Plywood
PP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-S-4473	Shielding, Magnetron Tubes and Magnets for Air Shipment
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloy
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-W-22759	Wire, Electric, Fluorocarbon-Insulated, Copper and Copper Alloy
MIL-C-26524	Calibrator Set, Magnetic Compass, Type A/E37T-10

Standards

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment

FSC 6605

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MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS24264	Connector, Receptacle, Electrical Flange Mount, Miniature
MS24266	Connector, Plug, Electrical - Straight, Miniature

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The detectors furnished under this specification shall be products that are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.5).

3.2 Components. The detector shall consist of the following:

a. Transmitter, Induction TRU-79/A

b. Compensator, Electronic MXU-394/A

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions encountered in storage or normal service. Metals shall also withstand the high temperatures specified herein.

3.4.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.4.3 Core material. The material for the magnetic core of the transmitter shall be such that it will withstand the maximum operating temperature specified herein plus the temperature rise due to heat dissipation from the electrical circuitry, without permanently altering the magnetic properties of the material.

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3.4.4 Wire. Wire for the electrical circuits of the transmitter shall be in accordance with MIL-W-22759 using silver-plated, class I conductors except that insulation material shall be teflon.

3.4.5 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, noninert materials shall be treated with a fungicidal agent acceptable to the procuring activity. However, if these materials are used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.6 Protective treatment. When materials are used in the construction of the detector that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of atmospheric conditions shall not be used.

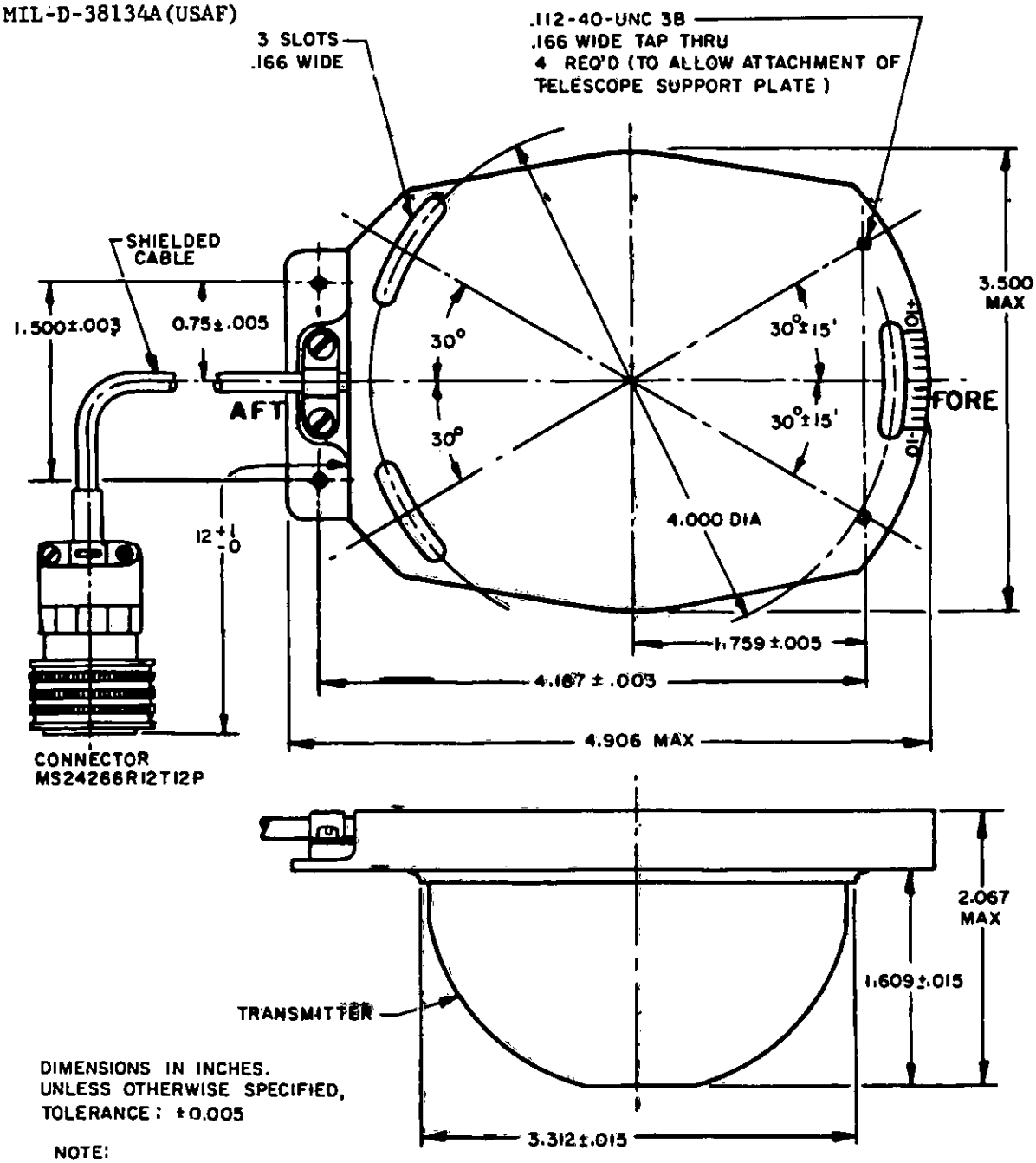
3.5 Design and construction. The detector shall be suitable for use with the A/E37T-10 magnetic compass calibrator set conforming to MIL-C-26524.

3.5.1 Transmitter. The transmitter shall conform to figure 1. It shall provide a minimum of 10° index error removal for either top or bottom mounting. When properly energized in the earth's magnetic field, the transmitter shall provide the indicator signal transformer with the proper direction signal for the compass system. The three secondary signal coils shall be internally Y-connected with the neutral point made externally available (see figure 2). The secondary coils leg-to-leg inductance when measured at 1,000 cycles with a standard laboratory instrument (such as a General Radio type 650-A impedance bridge or equivalent) shall be 92 ±10 millihenries and the d-c resistance shall be 30 ±3 ohms. When the primary coil is excited at 23.5V 400 Hz, the signal frequency of the secondary coils shall be 800 Hz and the impedance shall be between 500 and 800 ohms at an angle between 60° and 70°. The corresponding impedance of the secondary coils (leg-to-leg) shall be between 270 and 330 ohms at an angle between 72° and 88° at the 800-Hz signal frequency. The transmitter shall not become magnetized either by the presence of strong magnetic fields or by a d-c current passing through the secondary coils.

3.5.1.1 Mechanism. The transmitter mechanism shall be mounted pendulously with respect to the case.

3.5.1.2 Pendulous element. The earth inductor shall be mounted on a pendulous element so that it will tend to be horizontal when at rest. The pendulous element shall have at least 30° of freedom in pitch and at least 10° of freedom in roll with respect to the case. The pendulous element shall have a time period of not less than 0.2 second nor more than 0.75 second. The maximum permissible scale error shall be ±1.25° at a tilt angle of ±15° in pitch or ±10° in roll.

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NOTE:
AT THE OPTION OF THE PROCURING ACTIVITY, THE TWO AFT HOLES FOR ATTACHMENT OF THE SUPPORT PLATE MAY BE LOCATED ON A 3.812 ±0.003 DIAMETER AT 10° ± 15' RADIALS ON EACH SIDE OF THE FORE-AFT CENTERLINE.

FIGURE 1. TRU-79/A Transmitter

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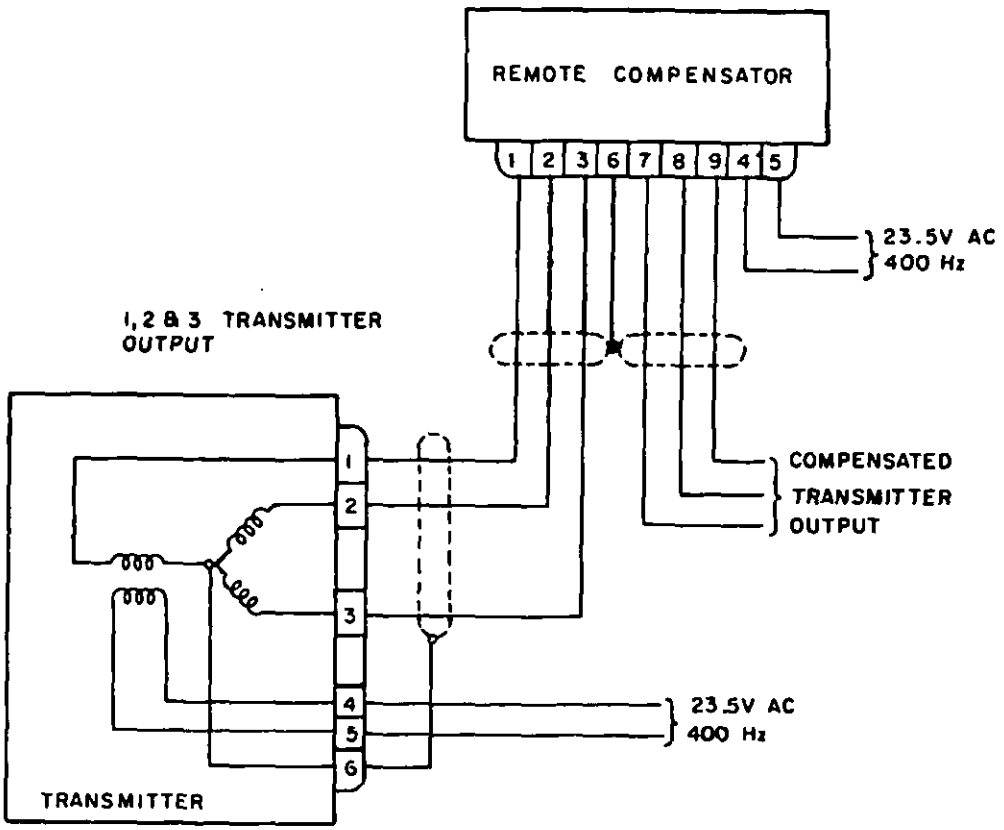


FIGURE 2. INTERCONNECTIONS

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3.5.1.2.1 Damping. Fluid-type damping shall be used to stabilize the pendulous element. The damping system shall permit not more than 12 oscillations to occur when the pendulous element is made to swing from any position up to its maximum degree of freedom.

3.5.1.3 Sealing. The transmitter shall be hermetically sealed to prevent leakage of the damping fluid.

3.5.2 Compensator

3.5.2.1 General design. The compensator shall be designed for mounting remotely from the transmitter. Interconnection between the compensator and transmitter shall be as shown on figure 2. The degree of compensation specified herein shall apply for interconnecting lead lengths between the compensator and transmitter of up to 200 feet of No. 22 wire. The compensator shall incorporate two screwdriver slot-type adjustments by means of which it shall be possible to compensate for errors in the heading signal produced in the transmitter. One adjustment shall be capable of compensating for north-south signal transmitter errors and shall be labeled N-S and the other adjustment shall be capable of compensating for east-west transmitter errors and shall be labeled E-W. The compensator shall conform to figure 3.

3.5.2.2 Neutral position influence. With both the N-S and E-W adjustments set on their neutral positions, the compensator shall not change the heading signal produced by the transmitter by more than 1° upon removal or reinsertion of the compensator into the transmitter circuitry.

3.5.2.3 Compensator range. With the transmitter set on 0° heading, the N-S adjustment shall be capable of changing the output signal by ±4° to ±6° for the first ±360° rotation of the N-S adjustment from its neutral position. An additional 360° rotation of the N-S adjustment in the same direction shall deflect the output signal at least 9° from the neutral position. With the transmitter setting on 90°, the E-W heading shall be capable of changing the output signal by ±4° to ±6° for the first ±360° rotation of the E-W adjustment from the neutral position. An additional 360° rotation of the E-W adjustment in the same direction shall deflect the output signal to at least 9° from the neutral position. A clockwise rotation of each adjustment as specified above shall cause an increase in the resultant heading signal.

3.5.2.4 Compensator adjustment interference. A 360° rotation of the E-W adjustment screw shall not influence the transmitter output signal by more than ±0.5° when it is set at 0° heading. When set at 90° heading, a rotation of 360° on the N-S adjustment shall not influence the transmitter output signal by more than ±0.5°.

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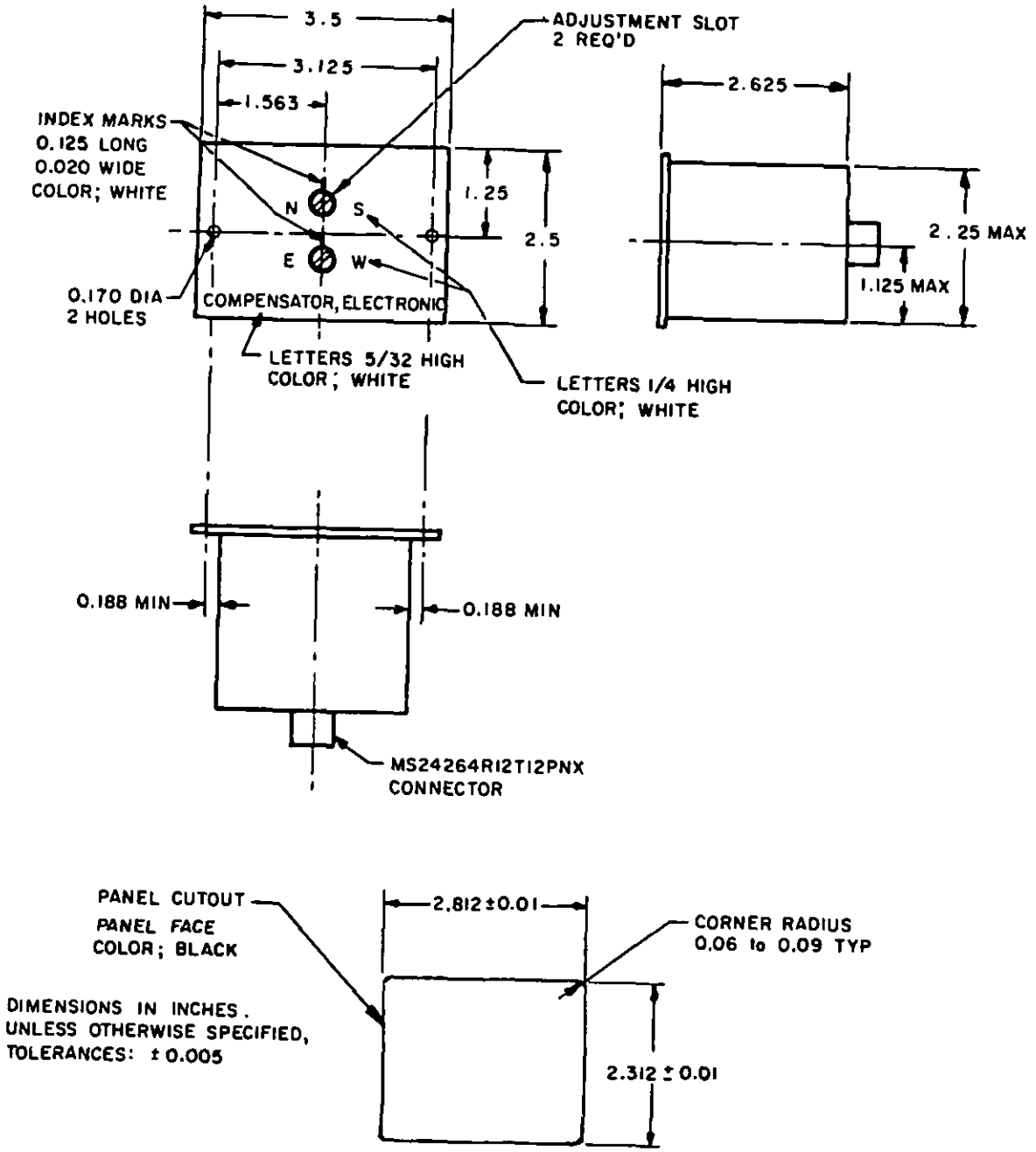


FIGURE 3 MXU-394/A COMPENSATOR

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3.5.3 Reliability. Reliability of the detector shall be a major design criterion. The detector shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,500 hours at a 90 percent confidence level. Failure shall be considered as leakage or any malfunction which relults in degraded performance regardless of the cause (careless handling or involuntary destruction excepted).

3.5.3.1 Longevity. The detector shall have a service life (longevity as defined in MIL-STD-781) of 5,000 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF). Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.5.4 Maintainability. The mean corrective maintenance downtime (M_{ct}) shall be 100 minutes not including the time required to unseal and reseal the detector and the maximum corrective maintenance downtime (M_{ctmax}) shall be not more than 300 minutes including the time required to unseal and reseal the detector.

3.6 Performance. The detector shall be capable of operating as specified herein after subjection to the following conditions:

a. Fungus - fungus as encountered in tropical climates

b. Dust - sand and dust particles as encountered in desert areas

c. Humidity - relative humidity up to 95 percent

d. Salt fog - salt sea atmosphere for 48 hours

e. Shock - shock of 20g for 11 milliseconds

3.6.1 High temperature. The transmitter shall be capable of operating continuously in ambient temperatures up to 200°C, and the compensator shall be capable of operating continuously in ambient temperatures up to 71°C.

3.6.2 Low temperature. The detector shall be capable of operating continuously in ambient temperatures down to -55°C.

3.6.3 Vibration. The detector shall be capable of operating as specified herein under all vibratory conditions incident to normal use with not more than ±1° signal error.

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3.6.4 Voltage and frequency variation. A shift in the input voltage and frequency to any combination between 22 and 25V and 360 and 440 Hz shall not cause the heading signal to shift more than ±0.5° at any point on the azimuth.

3.6.5 Altitude. The detector shall operate at altitudes up to 80,000 feet at any operational temperature range and within the tolerances for the ranges specified herein.

3.6.6 Explosive atmosphere. When operated in an explosive atmosphere, the compensator shall not cause an explosion.

3.6.7 Acceleration. The detector shall withstand an acceleration of 3g during continuous operation and the following accelerations in the directions specified:

a. Fore: 9.0g

b. Aft: 3.0g

c. Up: 4.5g

d. Down: 13.5g

e. Lateral: 6.0g.

3.6.8 Level field. Variation of the vertical field from 0.54 ±0.02 to less than 0.02 oersted shall not change the scale error output of the transmitter by more than ±0.3°.

3.6.9 Signal harmonic components. When tested as specified in 4.6.11, the maximum 400-Hz signal shall not exceed 25 mv and the maximum 1,200-Hz signal shall not exceed 40 mv.

3.6.10 Accuracy. Accuracy of the transmitter under normal operating conditions and at temperatures from -30° to +150°C shall be within ±0.4° of true heading; between 150° and 200°C shall be within ±0.8° of true heading; and between -55° and -30°C shall be within ±1.25° of true heading.

3.6.11 Sensitivity. With the transmitter set to the 0° or north heading and the signal transformer set to a heading of 90° ±0.05° greater than the transmitter, a current application of 1.07 ±0.05 ma from pins 1 to 6 shall result in a null indication on the signal transformer output.

3.6.12 Crosstalk error (180°). When the transmitter is set at the 0° or north heading, a current application of 2.14 ±0.1 ma dc from pin 1 to 6 shall generate a 180° electrical heading. The error at this heading shall not differ from the corresponding scale error at 180° by more than ±0.8°.

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3.6.13 Crosstalk error (270°). When the transmitter is set on the 90° heading, a current application of 1.2 ±0.1 ma dc from pin 2 to pin 3 shall generate a 270° electrical heading. The error at this heading shall not differ from the corresponding scale error at 270° by more than ±0.8°.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Power

3.8.1 Input. Excitation for the primary circuit of the detector shall be 23.5 ±2.0V 400 ±40-Hz single-phase a-c power.

3.8.2 Output. The 800-Hz signal output from the transmitter secondary circuit shall be transmitted over a 3-wire line so that it may be connected to the stator of the signal transformer.

3.8.3 Maximum output. The transmitter and signal transformer shall be placed on the 0° heading (null position) and the 800-Hz signal voltages across the 1-2 and 1-3 detector legs shall be recorded. The transmitter and signal transformer shall then be rotated to the 30° heading (null position). The voltage across the 1-2 leg shall increase and across the 1-3 leg shall decrease. With the transmitter still on the 30° heading, the signal transformer shall be rotated to the 120° heading. The line-to-line voltage across the 1-2 leg shall be 140 ±25 mv.

3.9 Screw threads. Unless otherwise specified, screw threads shall conform to MIL-S-7742.

3.10 Electric soldering. Electric soldering shall be in accordance with MIL-STD-454, requirement 5.

3.11 Pivots, bearings, and gears. Pivots, bearings, and gears shall not bind in operation. These parts and parts necessary for electrical commutation (i.e, sliprings, brushes, and rotary contacts) shall be as nearly frictionless as possible except where friction is required.

3.12 Finishes and protective coatings

3.12.1 Aluminum-alloy parts. Aluminum parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.12.1.1 Small holes and case inserts need not be anodized.

3.12.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

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3.12.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.12.1.4 Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.12.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.12.2 Cadmium plating. Cadmium plating shall be in accordance with type I, class 3 of QQ-P-416.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Workmanship. The detector, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.14.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the detector, they shall be held or limited accordingly.

3.14.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.14.3 Cleaning. The detector shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign material during and after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of detectors shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

4.3.2 Attitude. Unless otherwise specified, the detector shall be tested in its normal operating position.

4.3.3 Vibration. Unless otherwise specified, the detector shall be subjected to vibration at a frequency from 25 to 35 Hz so that a point on the transmitter will describe, in a vertical plane, a circle with a diameter of not less than 0.002 nor more than 0.005 inch.

4.3.4 Supply voltage. Unless otherwise specified, all tests shall be conducted with 23.5 ±0.5V 400 ±5-Hz single-phase a-c power applied to the detector.

4.3.5 Field strength. Unless otherwise specified, all tests shall be conducted with a horizontal field strength of 0.18 ±0.02 oersted and a vertical field strength of 0.54 ±0.02 oersted.

4.3.6 Standard signal transformer. All tests shall be conducted with a standard signal transformer having the following characteristics:

4.3.6.1 The signal transformer shall consist of a three-phase, three-wire stator surrounding a single-phase two-wire rotor. The maximum input line-to-line impedance of the stator with the rotor circuit open shall be 150 ±15 ohms at a phase angle of 82° ±3° when measured at 1,000 Hz with a standard laboratory instrument, such as a General Radio type 650-A impedance bridge, or equivalent.

4.3.6.2 The rotor impedance with the stator open shall be 10,000 ±1,000 ohms at a phase angle of 85° ±2°. When the rotor is tuned for maximum 800 Hz signal and 90 ±2 mv, 800 ±5 Hz is applied between one stator lead and a bus connecting the other two stator leads. The maximum 800 Hz signal from the rotor shall be 900 ±90 mv. The rotor shall be rotated until maximum voltage is obtained.

4.3.6.3 The signal transformer errors shall be minimum and known so that the error can be removed from the heading error during testing of the detector.

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4.3.6.4 The signal transformer shall be suitably mounted so that angular displacement from 0° to 360° may be determined within ±0.1°

4.3.7 Turntable. All tests shall be conducted with the detector mounted on a turntable accurately calibrated in increments of 15° from 0° through 360°.

4.3.8 Standard null meter. The standard null meter shall have a minimum input impedance of 1 migohm and shall be capable of detecting the null position of the standard signal transformer to within ±0.05° when the detector is in a horizontal field strength of 0.08 oersted and zero vertical field strength.

4.3.9 Romote compensator. Unless otherwise specified, the remote compensator shall be connected into the transmitter circuitry during all testing.

4.3.10 Magnetic heading simulator (MHS). A MHS may be used in lieu of the transmitter in the tests specified in 4.6.4 through 4.6.6. The MHS shall consist of a transmitter sensing element mounted in a double shield can and circuitry capable of controlling a magnetic field within the can. The MHS shall produce accurate and stable simulated magnetic heading in 15° increments from 0° to 360°.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of nine detectors representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Three samples shall be subjected to all the tests except reliability and the other six shall be subjected to the reliability test only.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.3).

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6 except 4.6.25.2.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each detector shall be subjected to the following tests as described under 4.6:

a. Transmitter

(1) Examination of product
(2) Scale error
(3) Sensitivity and cross talk tests

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(4) Level field
(5) Tilt error
(6) Damping
(7) Signal strength
(8) Signal harmonic component
(9) Magnetization.

b. Compensator

(1) Examination of product
(2) Neutral position
(3) Compensator range
(4) Interference.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One detector shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests in the sequence specified and as described under 4.6. These tests shall be completed prior to shipment of the group from which the detectors are selected.

a. Transmitter

(1) Individual tests
(2) Voltage and frequency variation
(3) Low temperature
(4) High temperature
(5) Temperature altitude
(6) Vibration failure
(7) Vibration error.

b. Compensator

(1) Individual tests
(2) Voltage and frequency variation
(3) Low temperature
(4) High temperature
(5) Temperature altitude
(6) Vibration failure
(7) Vibration error.

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three detectors shall be selected at random from the first 15 items of the contract or order and

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subjected to the following test in the order specified and as described under 4.6. These tests shall be completed within 75 days after the test units have been selected.

a. Sampling plan A tests
b. Dust
c. Acceleration
d. Explosive atmosphere (compensator only)
e. Shock
f. Special vibration (transmitter only)
g. Humidity
h. Fungus
i. Salt fog.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of detectors on hand or produced later shall not be made until it is determined that detectors meet all the requirements of the specification.

4.5.2.4 Sampling plan C. Unless otherwise specified (see 6.2), six detectors shall be selected at random from the first 20 units on each production run and subjected to the test specified in 4.6.25.2. Rejection shall be as specified therein.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in detectors already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The detector shall be examined to determine compliance with the requirements specified herein with respect to materials, design and construction, mechanical measurements, workmanship, and marking.

4.6.2 Scale error. The transmitter shall be mounted on a turntable in the normal position with standard field strength and standard input voltage. The transmitter secondary circuit shall be connected to the stator of a standard signal transformer whose rotor is connected to a standard null meter. The transmitter and signal transformer shall be set at the 0° or north heading (rotor at null position). The transmitter shall then be turned in heading increments of 30° through 360°. At each heading, the pointer of the signal transformer shall be

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turned correspondingly to its null position and the reading taken of the pointer on the dial. The difference between the pointer reading and the transmitter heading shall not exceed ±0.4° after the index error has been subtracted.

4.6.3 Sensitivity and cross talk tests. The following tests shall be performed with the transmitter connected to the signal transformer as specified in 4.6.2:

4.6.3.1 Sensitivity. The transmitter shall be set to the 0° or north heading and the signal transformer set to a heading of 90° ±0.05° greater than the transmitter. A direct current shall be applied from pins 1 to 6 and varied until a null is indicated on the standard null meter. The direct current shall be 1.07 ±0.05 ma.

4.6.3.2 Cross talk (180°). The transmitter shall be set at the 0° heading and a direct current of 2.14 ±0.1 ma shall be applied from pins 1 to 6. This shall produce a 180° heading in the transmitter. The error at this heading shall not differ from the error obtained in the scale error test at the 180° heading by more than ±0.8° when corrected for misalignment of the transmitter and readout.

4.6.3.3 Cross talk (270°). The transmitter shall be set at a 90° heading and a direct current of 1.2 ±0.1 ma shall be applied from pin 3 to pin 2. This shall produce a 270° heading in the transmitter. The error at this heading shall not differ from the error obtained in the scale error test at the 270° heading by more than ±0.8° when corrected for misalignment of the transmitter and readout.

4.6.4 Neutral position. The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator, with both north-south and east-west screws set at neutral, shall then be connected to the transmitter circuitry. The resulting shift of the null position of the signal transformer rotor shall not exceed ±1°. This test shall be repeated with the transmitter set on the 90° heading.

4.6.5 Compensator range. The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator, with both north-south and east-west screws set at neutral, shall be connected to the transmitter circuitry. The north-south screw shall then be turned 360°. The shift in null position of the rotor shall be between 4° and 6°, plus or minus, depending on the direction of rotation of the adjustment screw. The same screw shall then be turned an additional 360° in the same direction. The resulting shift in null position of the rotor shall be at least 9° from the neutral position. This test shall be repeated with the transmitter set on the 90° heading using the east-west screw and leaving the north-south screw at neutral. The corresponding shift in null position shall be within the limits specified herein for the north-south adjustment range.

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4.6.6 Interference. The transmitter shall be set at the 0° heading as specified in 4.6.2. The compensator shall then be connected to the transmitter circuitry with the north-south screw set at neutral and the east-west screw set at 360° from neutral. The resulting shift in null position of the rotor shall not exceed ±0.5°. This test shall be repeated with the transmitter set on the 90° heading, the east-west screw set at neutral, and with the north-south screw set 360° from the neutral position. The resulting shift in null position of the rotor shall not exceed ±0.50.

4.6.7 Level field. The transmitter shall be connected to the signal transformer as specified in 4.6.2 on the 0° heading. The horizontal field strength shall be 0.18 oersted and the vertical field shall be less than 0.02 oersted. Heading errors shall be recorded in increments of 60° through 360°. The difference between these after index error has been subtracted shall not exceed ±0.3°.

4.6.8 Tilt error. The transmitter shall be connected to the signal transformer as specified in 4.6.2 on the 0° heading. The transmitter shall be tilted up 15° in pitch and a scale error reading taken. This test shall be repeated with the transmitter tilted down 15° in pitch and then at a tilt of ±10° in roll. The error between the signal transformer rotor and the transmitter shall not exceed 1.25° at the above test points after the index error has been subtracted.

4.6.9 Damping. The transmitter shall be tilted rapidly from the level position to 10° in roll and return to the level position. The null position of the standard signal transmitter shall not oscillate more than 0.1° after 12 oscillations. This test shall be repeated except the transmitter shall be tilted 15° in pitch.

4.6.10 Signal strength. The detector shall be connected to the signal transformer as specified in the scale error test on the 0° heading. The signal transformer then shall be rotated to its 90° heading and the 800-Hz voltage across the rotor leads recorded. The voltage shall be not less than 1,000 mv.

4.6.11 Signal harmonic components. The detector shall be connected to the signal transformer as specified in the scale error test. The signal transformer shall be rotated for a maximum 400-Hz signal which shall not exceed 25 mv. It shall then be rotated for a maximum 1,200-Hz signal which shall not exceed 40 mv.

4.6.12 Magnetization. The transmitter shall be set at the 0° heading with field strengths as specified in the level field test. A d-c current of 100 ±10 ma shall be passed through the transmitter secondary coils in turn, 1 to 6, 2 to 6, and 3 to 6. Heading errors shall be recorded in increments of 90°. The difference between these errors, when corrected for index error, and the corresponding errors obtained in the level field test shall not exceed 0.2°. This test shall be repeated with the d-c current passed through the secondary coils in the 6 to 1, 6 to 2, and 6 to 3 direction.

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4.6.13 Voltage and frequency variation. The detector shall be set up as specified in the scale error test with the signal transformer on the 0° heading. The supply voltage shall then be reduced to 22V and the frequency to 360 Hz. The shift in null position of the signal transformer rotor shall not exceed ±0.5°. This test shall be repeated on two other headings not less than 100° apart. The supply voltage shall then be increased to 25V and the frequency to 440 Hz. The same test limits shall apply.

4.6.14 Low temperature. While inoperative, the detector shall be subjected to a temperature of -55°C for a period of 4 hours. At the end of this time and while at the specified temperature, it shall be energized and subjected to the scale error test, except that the tolerance shall be ±1.25°. The transmitter shall show no signs of leakage and the compensator shall operate without excessive binding or excessive freedom. The detector shall then be returned to room temperature. The transmitter shall meet the test specified in 4.6.2 and the compensator shall meet the test specified in 4.6.5.

4.6.15 High temperature. The transmitter and compensator shall be excited for a period of 4 hours at ambient temperatures of 200° and 71°C, respectively. At the end of the 4-hour period and while at the specified temperatures, they shall be subjected to the scale error test except that the tolerance shall be ±0.8°. The transmitter shall show no evidence of leakage and the compensator shall operate without excessive binding or freedom. The transmitter and compensator shall then be returned to room temperature and subjected to the tests specified in 4.6.2 and 4.6.5 respectively.

4.6.16 Temperature-altitude. The temperatures-altitude test shall be conducted as follows:

a. Step 1: The detector shall be subjected to a pressure equivalent to 80,000 ±1,000 feet at a temperature of -55° ±2°C for a continuous period of 3 hours. While operating at this pressure and temperature, the detector shall be subjected to the scale error test. The permissible scale error tolerance shall not exceed ±1.25° of true heading. This test shall be conducted on the 0° heading and two other headings not less than 100° apart. At the end of the 3-hour period, the detector shall show no sign of leakage nor shall there be any damage that would affect subsequent operation.

b. Step 2: The compensator only shall be removed from the chamber. A suitable connection between the transmitter and compensator shall be made through the chamber wall. The transmitter shall be then subjected to an altitude of 60,000 ±1,000 feet and a temperature of 160° ±2°C for a continuous period of 3 hours. While operating at this temperature and pressure, the scale error test shall be performed as in step 1 except the tolerance shall be ±0.8° of true heading. At the edd of the 3-hour period, there shall be no evidence of leakage or damage that would affect subsequent operation. The tests specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6 shall be conducted before and after this test.

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4.6.17 Vibration

4.6.17.1 Vibration failure (compensator). The compensator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, procedure I, figure 514.1-2, curve J. At the completion of this test, the compensator settings shall not be changed nor shall there be any looseness or damage. The compensator shall then be subjected to the individual tests.

4.6.17.2 Vibration failure (transmitter). While operating, the transmitter shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, procedure I, except the vibration curve shall be in accordance with figure 4. At the completion of this test, the transmitter shall be subjected to the individual tests. The transmitter shall show no evidence of leakage.

4.6.17.2.1 Special vibration test. The transmitter shall be vibrated at a frequency of 900 cpm with a total excursion of 0.125 inch in a horizontal plane for a period of 100 hours. The transmitter shall be mounted on the vibration machine in such a manner that the zero index line on the transmitter flange will be 45° from the direction of the vibration machine motion. At the completion of 20, 50, and 100 hours, the transmitter shall meet the tests specified in 4.6.2 and 4.6.8.

4.6.17.3 Vibration error. While operating, the detector shall be subjected to a vibration error test in accordance with MIL-STD-810, method 514, procedure I, figure 514.1-2 curve C, except the following test periods shall apply:

a. Resonance dwell - none

b. Sinusoidal cycling - 1 hour

c. Sweep time - 20 minutes.

During this test, the transmitter and compensator shall be interconnected. During vibration, the scale error shall be measured on the 0° heading and on two other headings not less than 100° apart. Scale error shall not exceed 1° of true heading.

4.6.18 Dust. The detector shall be subjected to a dust test in accordance with MIL-STD-810, method 510. At the end of this test, the detector shall be removed from the test chamber and allowed to cool at room temperature. The detector shall not be damaged. The compensator adjusting screws shall operate without excessive looseness. The detector shall then be subjected to the tests specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6.

4.6.19 Acceleration. The detector shall be subjected to an acceleration test in accordance with MIL-STD-810, method 513, procedure I. The transmitter and compensator shall then be subjected to the tests specified in 4.6.2, 4.6.4, 4.6.7, 4.6.8, and 4.6.9, as applicable.

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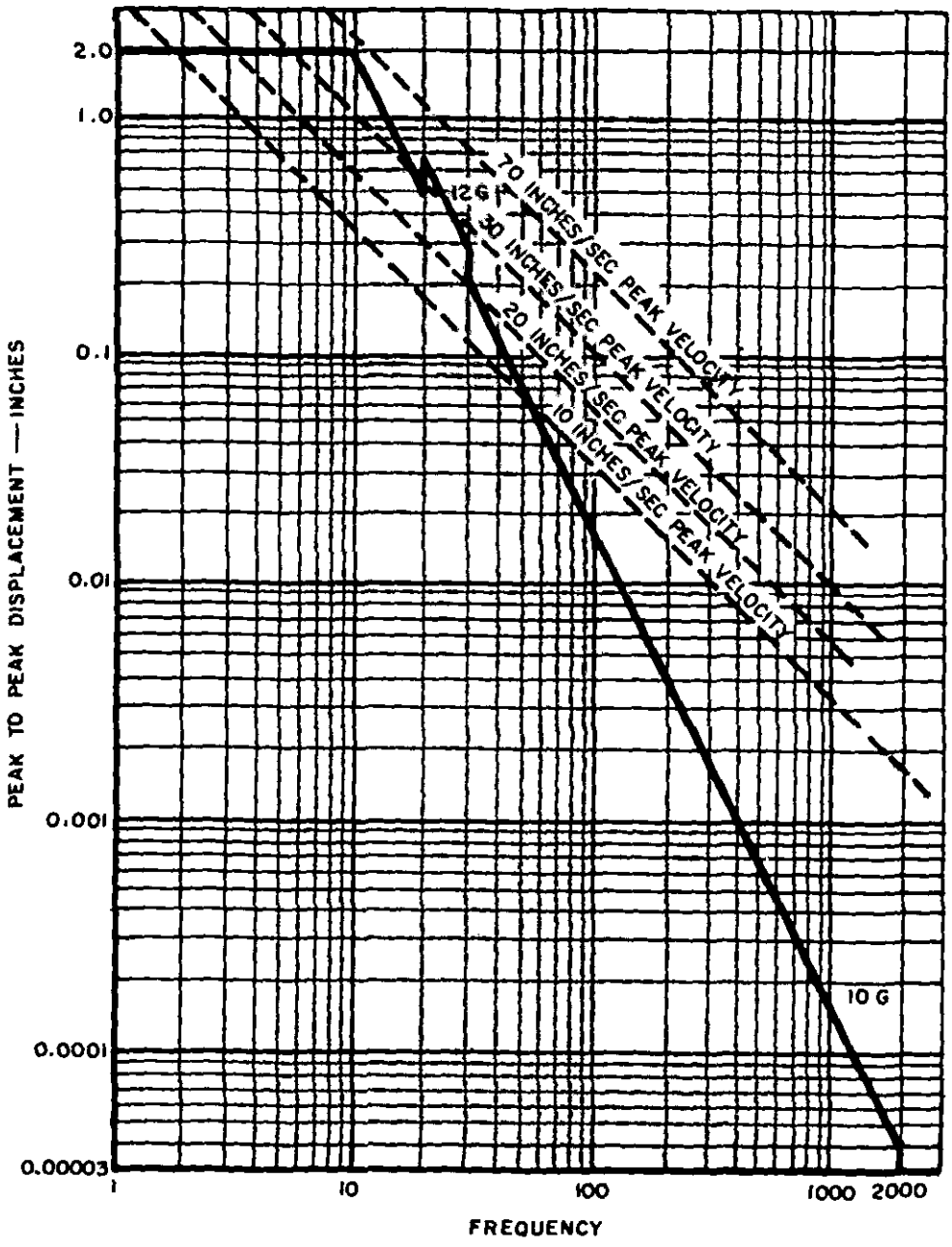


FIGURE 4. Vibration

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4.6.20 Explosive atmosphere. The compensator shall be subjected to an explosion-proof test in accordance with MIL-STD-810, method 511, procedure I.

4.6.21 Shock. The detector shall be subjected to a shock test in accordance with MIL-STD-810 method 516, procedure I, figure 516.1-1, amplitude (a) and time duration (c). At the completion of this test the detector shall show no signs of leakage or damage that would affect subsequent operation. The detector shall then be subjected to the tests specified in 4.6.2 and 4.6.4 through 4.6.9.

4.6.22 Humidity. The detector shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I. The detector shall then be subjected to the tests specified in 4.6.2, 4.6.4 through 4.6.6, 4.6.10, and 4.6.11. There shall be no evidence of corrosion or rust of internal or external parts.

4.6.23 Fungus. The detector shall be subjected to a fungus test in accordance with MIL-STD-810, method 508. At the conclusion of this test, the detector shall be subjected to the tests specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6. There shall be no evidence of deterioration that might adversly affect subsequent operation nor shall fungus growth be supported.

4.6.24 Salt fog. The detector shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509. At the conclusion of this test, the detector shall be subjected to the tests specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6. The compensator adjusting screws shall operate without excessive sticking, binding, or looseness. There shall be no evidence of corrosion of internal or external parts.

4.6.25 Reliability

4.6.25.1 Qualification. The six detectors specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level J, test plan III, except as follows:

a. Temperatures: Temperatures for the transmitter shall range from -54° to +200°C and for the compensator shall range from -54° to +71°C

b. Vibration: In addition to the vibration specified, the transmitter shall be subjected to a 2.0-inch double-amplitude vibration at any nonresonant frequency between 1 and 10 Hz for 50 minutes after every 100 hours of operation.

4.6.25.1.1 Each detector shall be operated a minimum of 1,500 hours. At the completion of every 250 ±10 hours of operation, the detector shall be subjected to the test specified in 4.6.2. At the completion of 500 ±20 hours of operation and at the conclusion of this test, the detector shall be subjected to all individual tests. If an accept decision can not be reached after a total accumulation of 20,100 hours of operating time, testing shall be stopped, the detector modified to eliminate the cause of failure, and the test shall then be repeated.

4.6.25.2 Quality conformance. The six detectors specified in 4.5.2.4 shall be subjected to a reliability test in accordance with 4.6.25.1 except the accept-reject criteria shall be in accordance with test plan V.

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4.6.26 Longevity. Unless otherwise specified, two of the detectors specified in 4.6.25.1 shall continue to be tested for a total of 10,000 hours (5,000 hours each). Evaluation of this test shall be in accordance with MIL-STD-781, test plan XXVIII.

4.6.27 Maintainability. The detector shall be subjected to a maintainability test in accordance with MIL-STD-471, appendix B, test method 1, test plan A_1 plus B_1.

4.7 Inspection of preparation for delivery. Preservation, packaging, and packing shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A (reference definition in MIL-STD-794). Unless otherwise specified in the contract or order, individual components of the detector shall be packaged one each in accordance with method IIC of MIL-P-116. Overbox complete pack in PPP-B-636 container utilizing cushioning of a type, density, configuration and thickness to insure shock, experienced by item does not exceed peak values established for the item, specified in g's, when completed packs are submitted to the rough handling tests of MIL-P-116.

5.1.2 Level C. Unless otherwise specified by the contract or order, individual components of the detector shall be packaged identically to level A requirements stipulated in 5.1.1 except method III of MIL-P-116 shall apply.

5.1.3 Magnetic shielding. Shielding as required shall be applied in accordance with MIL-S-4473.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Detectors preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Detectors preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

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5.2.3 Packing shall be applied which affords adequate protection during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

6. Notes

6.1 Intended use. The DSU-4/A detector covered by this specification is intended for use in airborne heading systems to detect the direction of the earth's magnetic field with respect to the direction of the aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title number and date of this specification

b. When sampling plan B or C tests will not be conducted

c. Levels of packaging and packing required.

6.3 Test report. The contractor will submit three copies of the test report specified in 4.4.2 to the procuring activity for approval.

6.4 Hermetic seal. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes brazing or soldering, or fusion by welding. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division Attn: ENFIF, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.6 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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